

Sri Lanka Institute of Information Technology



Web-based Employee Management System

2025-Y2-S1-MLB-B10G2-01

Student ID	Name
IT24102622	AMARASINGHE V. A
IT24102614	KAHADAWALA D. C
IT24102607	IRUSHI UMANDA M. M. A
IT24102627	FERNANDO W. D. N
IT24102669	SATHSARANI M. B. N. K
IT24102615	MAMALGAHA I. G. W. S

Software Engineering (SE2030)

B.Sc. (Hons) in Information Technology

Submission Date: _____

02) INTRODUCTION

Our project outlines the development of a Web-Based Employee Management System (EMS) for Lanka Tech Solutions (Pvt) Ltd. It is designed to make human tasks in an organization easier and more accurate. The EMS enables HR departments and managers to manage employee profiles, attendance, leaves, and shifts efficiently.

Objectives:

- Use digital methods to manage employee details and daily office work.
- To track employee attendance daily.
- To manage leave applications and approvals efficiently.
- Create a safe way to manage employee work and details.
- Make HR tasks easier and productive by reducing manual paperwork.

Target Stakeholders:

- HR Staff.
- Employees.
- Managers
- Admins.

Scope:

- Core modules are employee registration and profile management, attendance tracking, leave management, performance tracking, shift management and notification management.
- Admins can manage all employee data and system settings.

Limitations:

- Limited to web browser.
- Initially supports small to medium sized organizations.
- Requires a strong internet connection.
- Only Admins can access the system.
- Need to have basic knowledge to use the system.

03) REQUIREMENTS

Functional Requirements

1. Manage Employee Profiles

This function lets HR and admin staff add and update employee details such as adding new employees, editing employee information and removing employees when they leave.

2. Track Attendance

This helps record employee's daily attendance, View attendance and identify absentees.

3. Monitor Performance

This shows how well employees are engaged in their work by adding performance feedback, suggesting training and rating employee performance.

4. Manage Employee Shifts

This allows admin staff to create, update, and assign day and night shifts to employees, ensuring proper scheduling. It also enables employees to request and perform shift swaps with approval from admin staff.

5. Send Alerts and Notifications

This helps to remind employees and staff about poor attendance, missing updates and upcoming reviews

6. Manage User Roles and Access

This controls who can use what part of the system. It creates user accounts, assign roles, reset passwords and control access to features.

Non-functional Requirements

1) Performance

The system must respond quickly, with page loads and actions completed within 2-3 seconds. It should support at least 100 concurrent users without performance issues.

2) Usability

The interface must be simple and intuitive, allowing users with basic computer knowledge to easily perform tasks such as updating employee profiles or generating reports.

3) Reliability

The system should operate smoothly and be available 99.9% of the time during business hours. Regular backups will be implemented to protect data in case of system failure.

4) Security

User access will be controlled through role-based access control (RBAC). Passwords will be securely stored, and sensitive employee data will be encrypted. Password reset functionality will be available.

5) Maintainability

The codebase will follow Java best practices with a modular design, making it easy for future developers to update or enhance the system.

6) Compatibility

The system will work on modern browsers such as Chrome, Firefox, and Edge, and be accessible from both desktop and tablet devices.

7) Scalability

The system will be built with scalability in mind, allowing future expansion such as adding payroll features or increasing user capacity.

Constraints

1) Internet Dependency

The system is web-based and requires an active internet connection. It cannot function offline.

2) Platform Limitation

Access is restricted to desktop and tablet web browsers; no mobile application is included in this version.

3) Functional Scope

The system manages employee profiles, attendance, performance, and **shift scheduling**. Payroll processing and advanced leave management are not included in this release.

4) Security Limitations

Basic security features such as role-based access, password protection, and control over shift management are implemented. Advanced security measures, such as two-factor authentication or biometric login, are not included.

5) Automation Limitation

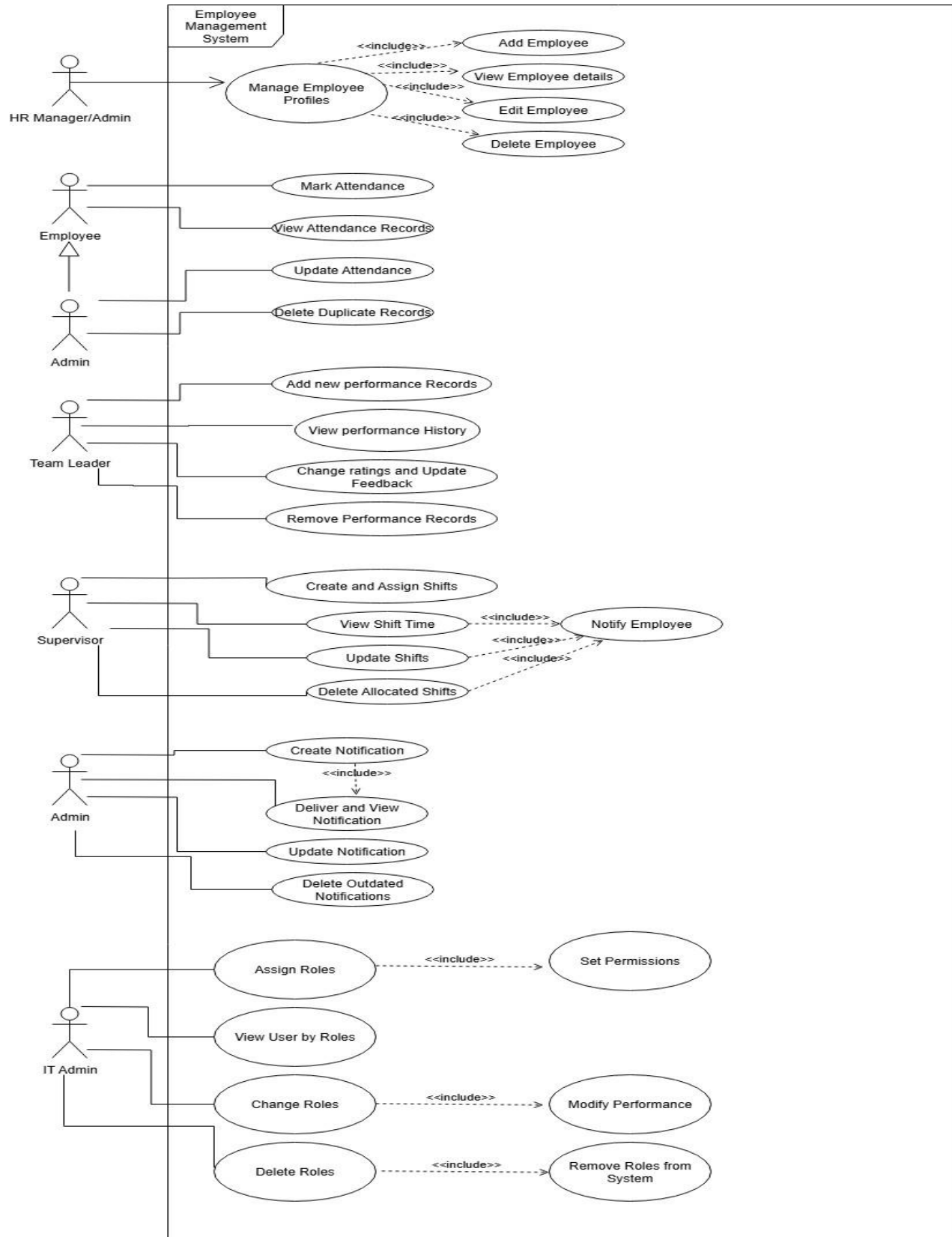
Attendance tracking, shift scheduling, and notifications operate on predefined rules; the system does not include AI-driven or intelligent automation.

6) Data Accuracy Dependency

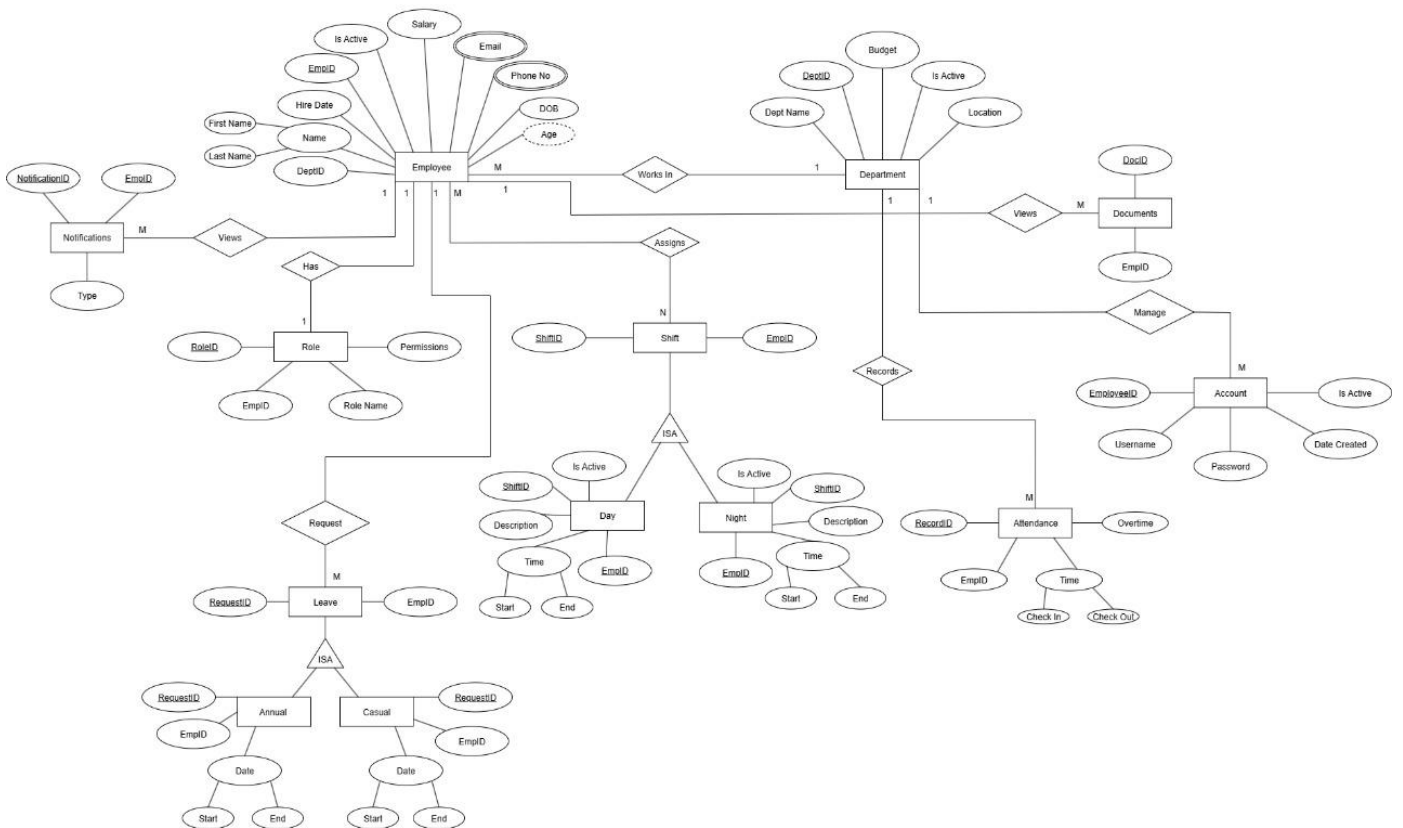
Accurate results for employee profiles, attendance, and shift management rely on correct and timely input by administrators.

04) DESIGN

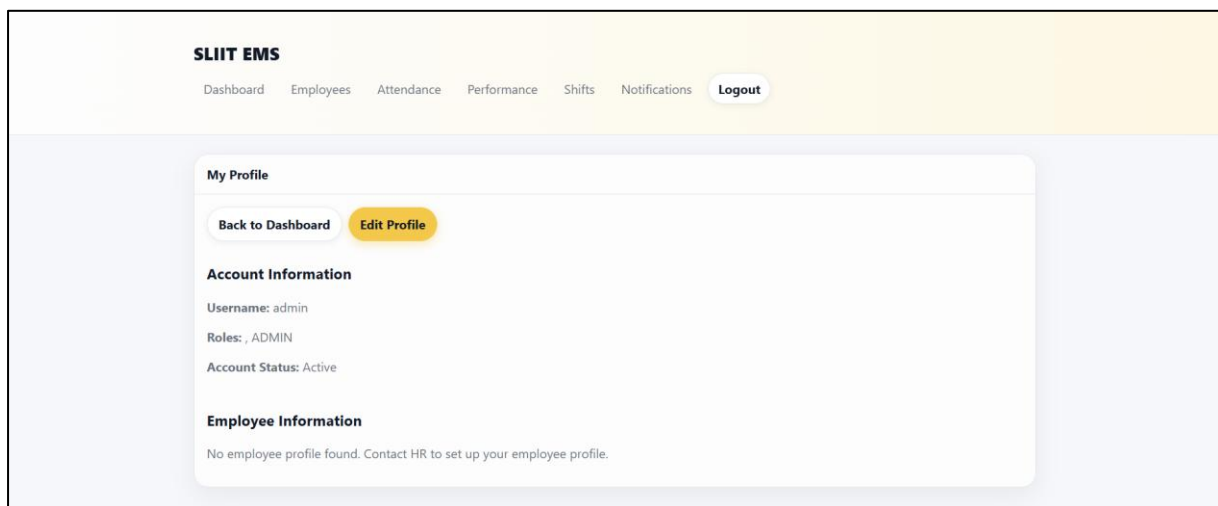
Use Case Diagram



Database Design



UI Screenshots



SLIIT EMS

MY AREA

My Dashboard

My Profile

My Attendance

My Shifts

Welcome, Tharushi

Overview

1

My Total Shifts

0

My Pending Swaps

My Upcoming Shifts

ID	TYPE	DATE	START	END
4	Night	2025-10-23	23:00	07:00

My Pending Swap Requests

ID	REQUESTED SHIFT	OFFERED SHIFT	STATUS
No pending swap requests			

EMS Login

Employee Management System

You have been logged out successfully.

Username

Enter your username

Password

Enter your password

Sign In

Demo Credentials

ADMIN

admin / admin123

HR

sadeni / sadeni123

MANAGER

nethmi / nethmi123

EMPLOYEE

tharushi / tharushi123

05) IMPLEMENTATION

Tools And Technologies Used

Category	Tool / Technology	Purpose / Usage
Programming Language	Java	Used as the main backend programming language to implement system logic, business processes, and data handling.
Database	SQL (MySQL)	Used to store and manage employee, attendance, and shift data securely and efficiently.
Integrated Development Environment (IDE)	IntelliJ IDEA	Used for writing, debugging, and testing the Java-based system efficiently.
Database Management Tool	SQL Server Management Studio (SSMS)	Used to manage and test SQL queries, visualize the database, and maintain data consistency.
Framework	Spring Boot	Used to build the backend web application and manage dependencies.
Design Tool	Figma	Used to design user interface wireframes and prototypes for the web application.
Communication & Collaboration Tool	Kyla	Used for task coordination, communication, and documentation during group collaboration.

Key Features Developed

- 1) Sathsarani M.B.N.K - Employee Profile Management
- 2) Mamalgaha I.G.W.S - Employee Shift Management
- 3) Amarasinghe V.A – Manage User Roles and Access
- 4) Fernando W.D.N – Monitor Performance
- 5) Irushi Umada M.M.A – Track Attendance
- 6) Kahandawala D.C – Send Notifications and Alerts

Screenshots of Core Functions

Login

EMS Login
Employee Management System

✓ You have been logged out successfully.

Username
Enter your username

Password
Enter your password

Sign In

Demo Credentials

ADMIN	admin / admin123
HR	sadeni / sadeni123
MANAGER	nethmi / nethmi123
EMPLOYEE	tharushi / tharushi123

[← Back to Home](#)

Employee Profile Management Page

Employees

Search name/email

All

Department

Job Title

Any Type

Filter

Add Employee

Back

ID	NAME	EMAIL	DEPT	TITLE	TYPE	ACTIONS
1	Admin User	db-admin@example.com	IT	System Administrator	Full-Time	View Edit Delete
2	HR Manager	db-hr@example.com	Human Resources	HR Manager	Full-Time	View Edit Delete
3	Manager User	db-manager@example.com	Operations	Operations Manager	Full-Time	View Edit Delete
4	Employee User	db-employee@example.com	IT	Software Developer	Full-Time	View Edit Delete
5	Dinaya Perera	dinaya@example.com	IT	Senior Developer	Full-Time	View Edit Delete
6	Sadeni Fernando	sadeni@example.com	Human Resources	HR Specialist	Full-Time	View Edit Delete
7	Nethmi Silva	nethmi@example.com	Operations	Team Lead	Full-Time	View Edit Delete

Employee Shift Management Page

SLIIT EMS

Dashboard

My Profile

Employees

Attendance

Calendar

Leave

Balances

Approvals

Performance

Shifts

Shift Calendar

Team Roster

Reports

Notifications

Logout

Shift Dashboard

Shift Calendar

Team Roster

Back to Dashboard

4

Total Shifts

2

Day Shifts

2

Night Shifts

0

Pending Swaps

Create New Shifts

Add Day Shift

EMPLOYEE:

Select Employee

DATE:

mm/dd/yyyy

START TIME (FIXED):

08:00 AM

END TIME (FIXED):

05:00 PM

Add Day Shift

Add Night Shift

EMPLOYEE:

Select Employee

DATE:

mm/dd/yyyy

START TIME (FIXED):

06:00 PM

END TIME (FIXED):

04:00 AM

Add Night Shift

Manage User Roles and Access Page

User Management

Back to Dashboard

Add User

ID	USERNAME	TYPE	STATUS	ROLES	LAST LOGIN	ACTIONS
1	db-admin	ADMIN	Active	ADMIN	Never	Edit Permissions Delete
2	db-hr	MANAGER	Active	HR	Never	Edit Permissions Delete
3	db-manager	MANAGER	Active	MANAGER	Never	Edit Permissions Delete
4	db-employee	EMPLOYEE	Active	EMPLOYEE	Never	Edit Permissions Delete
5	dinaya	ADMIN	Active	ADMIN	Never	Edit Permissions Delete
6	sadeni	MANAGER	Active	HR	Never	Edit Permissions Delete
7	nethmi	MANAGER	Active	MANAGER	Never	Edit Permissions Delete
8	tharushi	EMPLOYEE	Active	EMPLOYEE	Never	Edit Permissions Delete
9	hiruni	EMPLOYEE	Active	EMPLOYEE	Never	Edit Permissions Delete
10	admin	ADMIN	Active	ADMIN	Never	Edit Permissions Delete
11	hr	MANAGER	Active	HR	Never	Edit Permissions Delete

Monitor Performance Page

SLIIT EMS

Dashboard

My Profile

Employees

Attendance

Calendar

Leave

Balances

Approvals

Performance

Shifts

Reports

Notifications

Logout

Performance Reviews

Dashboard

Back to Dashboard

All employees

Filter

New Quarterly Review

Admin User

Q1 2025

Feedback

Add

New Annual Review

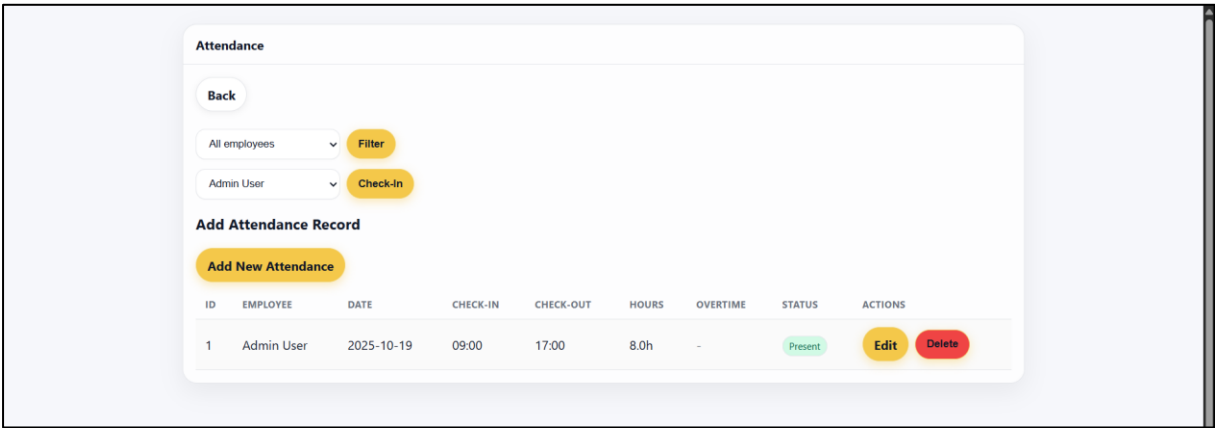
Admin User

Feedback

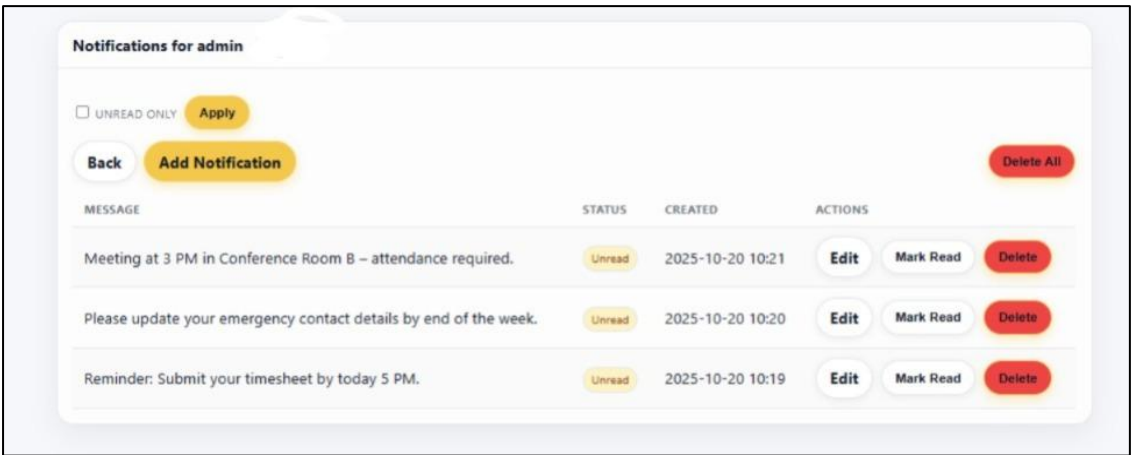
Add

ID	TYPE	EMPLOYEE	DATE	RATING	FEEDBACK	ACTIONS
1	Q1 2025	Admin User	2025-09-19	4/5	Strong performance in Q1	4 Strong performance in Q1 Update Delete
2	2024	Admin User	2025-04-02	5/5	Excellent year	5 Excellent year Update Delete

Track Attendance Page



Notifications and Alerts Page



06) PROJECT MANAGEMENT

Agile Approach & Sprint Summary

Sprint	Main Objectives	Key Tasks Completed	Outcomes
Sprint 1	Employee Profile & Shift Management	Implemented employee profile database; add/update functionality; shift CRUD operations; data validation rules for shifts.	Core employee and shift management modules completed and integrated with the database.
Sprint 2	Attendance & Alerts	Developed attendance tracking UI and backend; absentee alerts; reminders for missing employee details; contact info update.	Attendance management and automated alert system are fully functional.
Sprint 3	Performance & Feedback	Added performance feedback for employees; created team performance dashboard; linked attendance with performance trends.	Working performance monitoring and feedback modules for employees and team leaders.
Sprint 4	Analytics, Security & User Roles	Implemented dashboards; system security features; user account creation; role assignment and access control; shift summaries; training suggestions; goal setting; password reset.	Fully deployed system with security, analytics, and complete role-based access control.

Task Distribution

PBI ID	User Story	Priority	Sprint	Tasks	Assigned To
PBI-01	Add/update employee profiles	High	Sprint 1	Set up employee profile database structure; Implement add/update employee functionality	Sathsarani M.B.N.K
PBI-02	Manage Employee Shifts – CRUD	High	Sprint 1	Implement shift CRUD operations	Mamalgaha I.G.W.S
PBI-05	Manage Shift Data Validation	High	Sprint 1	Implement shift data validation rules	Mamalgaha I.G.W.S

PBI-03	View attendance records	High	Sprint 2	Implement attendance tracking UI and backend	Irushi Umanda M.M.A
PBI-04	Receive absentee alerts	Medium	Sprint 2	Configure absentee alert notifications	Irushi Umanda M.M.A
PBI-06	Receive reminders about missing employee details	Medium	Sprint 2	Implement reminders for missing employee details	Kahandawala D.C
PBI-07	Update employee contact information	Medium	Sprint 2	Enable admin to update employee contact info	Sathsarani M.B.NK
PBI-09	Receive performance feedback	Medium	Sprint 3	Implement individual performance feedback	Fernando W.D.N
PBI-10	Track performance trends	Medium	Sprint 3	Implement trend reports for engineers	Fernando W.D.N
PBI-13	View team performance stats	High	Sprint 3	Implement team performance dashboard for team leaders	Fernando W.D.N
PBI-14	Give feedback to team members	High	Sprint 3	Link attendance data with performance	Fernando W.D.N
PBI-16	Track attendance with performance	Medium	Sprint 3	Integrate attendance data into performance tracking	Irushi Umanda M.M.A
PBI-17	Access company-wide analytics	High	Sprint 4	Build dashboards for GM analytics	Kahandawala D.C
PBI-18	View and Analyze Shift Summaries	Medium	Sprint 4	Implement shift summary reports	Mamalgaha I.G.W.S
PBI-19	Secure system	High	Sprint 4	Implement system security features	Amarasinghe V.A
PBI-20	View real-time employee metrics	Medium	Sprint 4	Implement real-time metrics monitoring	Fernando W.D.N
PBI-21	Create/manage user accounts	High	Sprint 4	Implement user account management	Amarasinghe V.A
PBI-22	Assign roles	High	Sprint 4	Implement role assignment system	Amarasinghe V.A
PBI-24	Control feature access	High	Sprint 4	Implement feature access control	Amarasinghe V.A
PBI-08	Manage alerts for incomplete profiles	Low	Sprint 4	Implement alerts for incomplete profiles	Kahandawala D.C
PBI-11	Set personal goals	Low	Sprint 4	Enable employees to set personal goals	Sathsarani M.B.N.K
PBI-15	Suggest training sessions	Medium	Sprint 4	Implement training suggestions for team leaders	Fernando W.D.N
PBI-23	Reset passwords	Medium	Sprint 4	Add password reset functionality	Amarasinghe V.A

Development Phases & Timeline

Phase	Timeline	Key Milestones
Sprint 1	Weeks 1–3	Employee Profile and Shift Management modules completed.
Sprint 2	Weeks 4–6	Attendance and Notification modules developed and tested.
Sprint 3	Weeks 7–9	Performance Monitoring and Feedback systems implemented.
Sprint 4	Weeks 10–12	Analytics, Security, and User Role Management completed; Final integration and testing done.

07) Conclusion & Future Work

Summary of Achievements

The Employee Management System project was successfully designed and implemented to automate and simplify key human-resource operations. The system enables administrators, HR managers, and employees to manage information efficiently through an integrated and user-friendly web platform.

Major achievements include:

- ✓ Employee profile creation, update, and management
- ✓ Shift scheduling and validation
- ✓ Attendance tracking and absentee alerts
- ✓ Employee performance monitoring with quarterly and yearly ratings
- ✓ Automated notifications and reminders
- ✓ Role-based user access and permission control
- ✓ Company-wide analytics and reporting dashboards
- ✓ A secure, fully integrated system connecting all modules in real time

The project achieved its main objective of improving workforce management efficiency while reducing manual errors.

Challenges Faced

- Database Integration: Linking multiple modules to a centralized database initially caused inconsistencies.
- Shift and Schedule Conflicts: Designing validation logic to prevent overlapping schedules required careful rule creation.
- Performance Rating: Creating a fair evaluation model was difficult.
- Notification Automation: Implementing scheduled alerts and reminders required careful testing.
- Time and Coordination: Managing parallel development tasks and aligning team progress demanded effective sprint planning and communication.

Suggestions for Improvement

- **Mobile Application:** Develop a mobile version for employee self-service.
- **Advanced Analytics:** Add visual dashboards and data visualization for performance and attendance trends.
- **AI-Based Automation:** Use AI to recommend shift schedules, detect anomalies, and predict absentee patterns.
- **Biometric Integration:** Link attendance tracking with biometric or RFID systems for greater accuracy.
- **Cloud Deployment:** Host the system on a cloud platform to improve accessibility and scalability.
- **Enhanced Security:** Implement two-factor authentication.

08) INDIVIDUAL CONTRIBUTION

Member Name	Student ID	Role / Tasks	Challenges Faced	How Overcame Challenges	Key Lessons Learned	Reflection on Project
Sathsarani M.B.N.K	IT24102669	Employee Profile Management – designed and implemented employee profile module; handled add, edit, delete, and view functionalities; integrated with the database.	Encountered issues with form validation and SQL data insertion errors.	Researched JDBC connection handling and applied field-level validation techniques.	Learned database integration, validation, and backend debugging skills.	The module worked smoothly after testing; UI design and data filtering could be enhanced.
Mamalgaha I.G.W.S	IT24102615	Employee Shift Management – developed functionality to create, assign, and update employee shifts based on dates and roles.	Faced problems with overlapping shift assignments and date/time conflicts.	Implemented validation rules and logic checks to prevent conflicting schedules.	Improved logical thinking, error-handling, and scheduling algorithm design.	The module functioned effectively; it could be improved by adding auto-suggestions or shift templates.
Amarasinghe V.A	IT24102622	Manage User Roles & Permissions – created access levels for Admin, HR Manager, and Employees with appropriate rights.	Faced issues where access restrictions were not applied correctly.	Debugged session management and added conditional checks for each role.	Learned how to design secure systems and manage user access control effectively.	The feature increased system security; more testing could improve reliability.

Fernando W.D. N	IT24102627	Monitor Performance – developed performance evaluation module that rates employees quarterly and yearly based on attendance and shift completion.	Difficulty in defining fair rating criteria and ensuring accurate calculations.	Designed a weighted scoring model combining attendance, punctuality, and completed shifts after discussing with the team.	Learned how to create fair evaluation metrics and analyze performance data over time.	The module provided valuable performance insights; adding visual charts could make it more user-friendly.
IRUSHI UMANDA M.M.A	IT24102607	Track Attendance – implemented attendance tracking and absentee detection system connected with employee profiles.	Encountered duplicate records and missing attendance logs.	Added unique constraints and automated data validation checks to fix inconsistencies.	Gained experience in data validation, database constraints, and error prevention.	The system ran reliably; biometric or RFID integration could be a future enhancement.
KAHANDAWALA D.C	IT24102614	Send Notifications & Alerts – built automatic alert system to notify HR about absences, shift reminders, and performance alerts.	Faced difficulties with scheduling notifications and email integration.	Implemented timed schedulers and tested alert triggers with mock data.	Learned automation, asynchronous task scheduling, and system communication handling.	The feature improved system responsiveness: push notifications could be added in the future.

09) REFERENCES

[1] Oracle, “Java Platform, Standard Edition Documentation,” Oracle, 2024. [Online]. Available: <https://docs.oracle.com/javase/>

[2] MySQL, “MySQL Reference Manual,” Oracle Corporation, 2024. [Online]. Available: <https://dev.mysql.com/doc/>

[3] TutorialsPoint, “JDBC - Introduction,” 2024. [Online]. Available: <https://www.tutorialspoint.com/jdbc/index.htm>

[4] W3Schools, “HTML Tutorial,” 2024. [Online]. Available: <https://www.w3schools.com/html/>

[5] GeeksforGeeks, “Java Servlet Tutorial,” 2024. [Online]. Available: <https://www.geeksforgeeks.org/servlets-in-java/>

[7] Stack Overflow, “How to handle JDBC connection errors,” 2024. [Online]. Available: <https://stackoverflow.com>

10) APPENDIX

GitHub link

[https://github.com/IT24102622/EMS Repo](https://github.com/IT24102622/EMS_Repo)